Inicios de viviendas y tasa hipotecaria en EE.UU.

Trabajo práctico de Econometría II

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Resumen

- **Pregunta.** ¿Cuánto y con qué rezago caen los inicios de viviendas cuando sube la tasa hipotecaria? Causa: la tasa.
- **Datos.** Mensuales, 1985-01 a 2026-08 ($T = 500$), en $100\ln$.
- **Inicios.** $(1-B)(1-B^{12})$ y ARIMA$(0,1,1)(0,1,1)_{12}$ con atípico en 2020-04.
- **Tasa.** $(1-B)$ y ARIMA$(0,1,1)$.
- **Relación.** Espuria en niveles. Filtrada: elasticidad a 12 meses $-0{,}54$ (ee 0,33), unos $-7{,}6\,\%$ de inicios por $+1$ pp de tasa. Hay retroalimentación.

1. Objetivo y datos

Objetivo. Medir la respuesta de los inicios de viviendas a la tasa hipotecaria. La tasa es causa, no parte: encarece la cuota y se forma en el mercado de bonos.

Serie	Código FRED	Unidad	Frecuencia	Transformación
Inicios de viviendas privadas	HOUSTNSA	miles, sin desestacionalizar	mensual	$x_t = 100\ln$
Tasa hipotecaria fija 30 años	MORTGAGE30US	% anual	semanal $\to$ promedio mensual	$y_t = 100\ln$

Cuadro 1: Series. Muestra: 1985-01 a 2026-08 ($T = 500$).

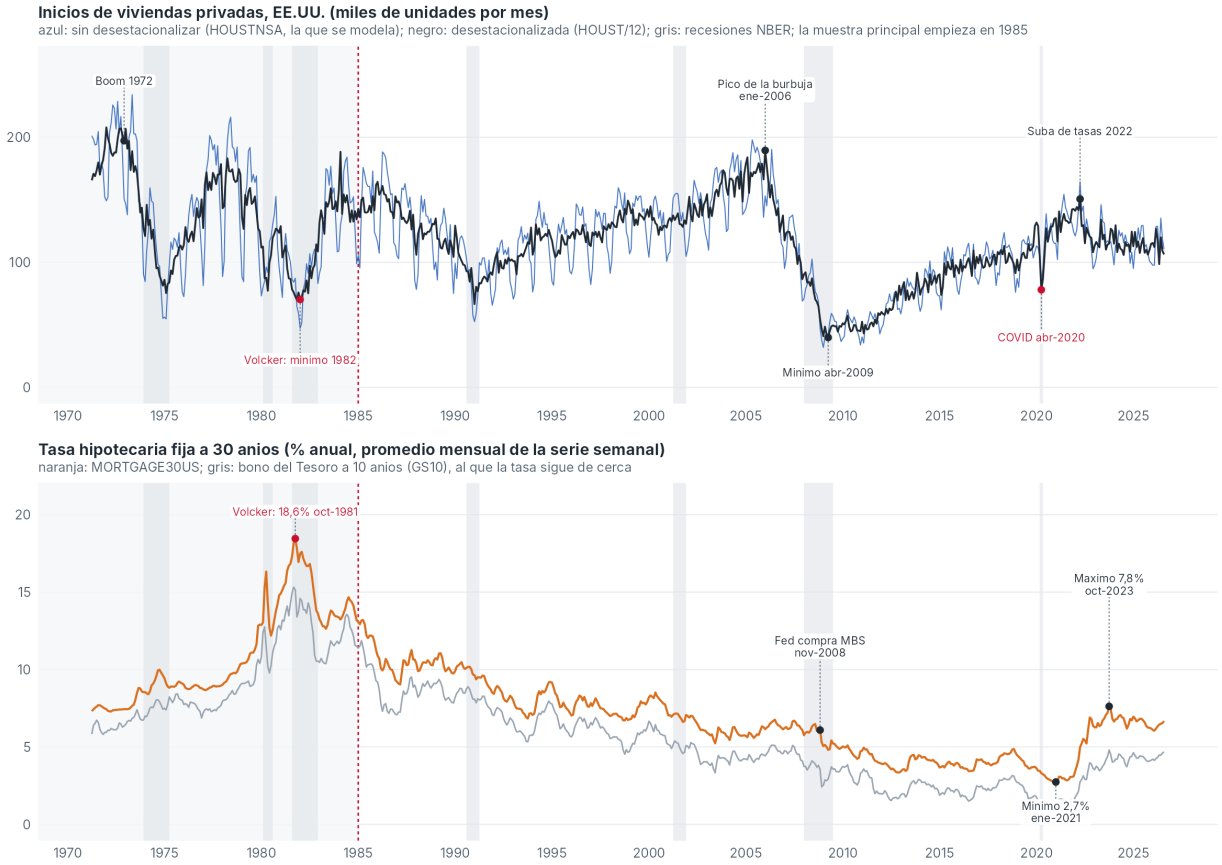


Figura 1: Series en niveles, 1971–2026. Gris: recesiones NBER.

Decisiones.

- **Muestra desde 1985.** En 1979–84 (Volcker) el desvío de Δtasa es 0,58 pp, 2,6 veces el de 1985–2007 (0,23 pp).
- **Serie sin desestacionalizar.** La estacionalidad se modela (Sección 4).
- **Atípico 2020-04** (cierre por COVID), tratado como AO.

Período	Desvío de Δ tasa (pp)
1971–1978	0,13
1979–1984 (Volcker)	0,58
1985–2007	0,23
2008–2019	0,16
2020–2026	0,25

Cuadro 2: Desvío de la variación mensual de la tasa.

2. Estacionalidad

Serie	Estac. fija	p	R^2 dummies	Cambio de patrón	p
Δ ln inicios	$F_{11,486} = 43{,}5$	$< 0{,}001$	0,50	$F_{11,475} = 4{,}13$	$< 0{,}001$
Δ ln tasa	$F_{11,486} = 1{,}37$	0,182	0,032	$F_{11,475} = 0{,}91$	0,532

Cuadro 3: Δ de cada serie sobre 11 dummies mensuales centradas. Cambio de patrón: mes $\times$ subperíodo (1985–2007 vs. 2008–2026).

- **Inicios:** estacionalidad fuerte ($F = 43{,}5$, $R^2 = 0{,}50$) y patrón que cambia ($F = 4{,}13$, $p < 0{,}001$) $\Rightarrow$ estocástica $\Rightarrow$ $(1 - B^{12})$ *(Nota 6, Sec. 2.3)*.
- **Tasa:** sin estacionalidad ($p = 0{,}182$).

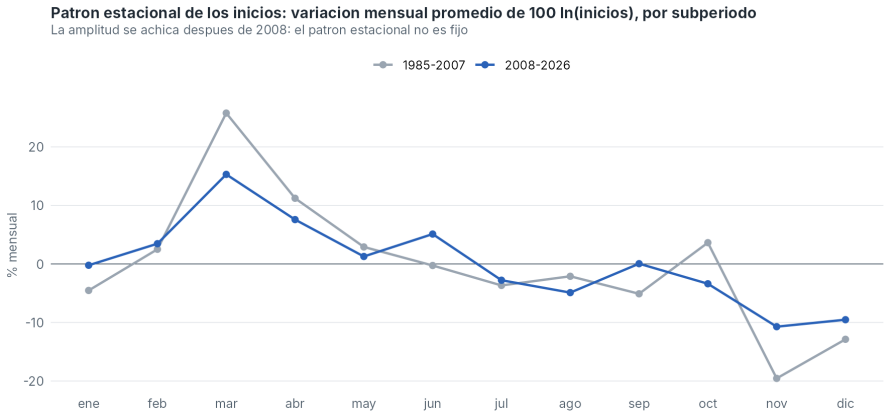


Figura 2: Variación mensual promedio de los inicios por subperíodo.

3. Estacionariedad

3.1. Correlogramas

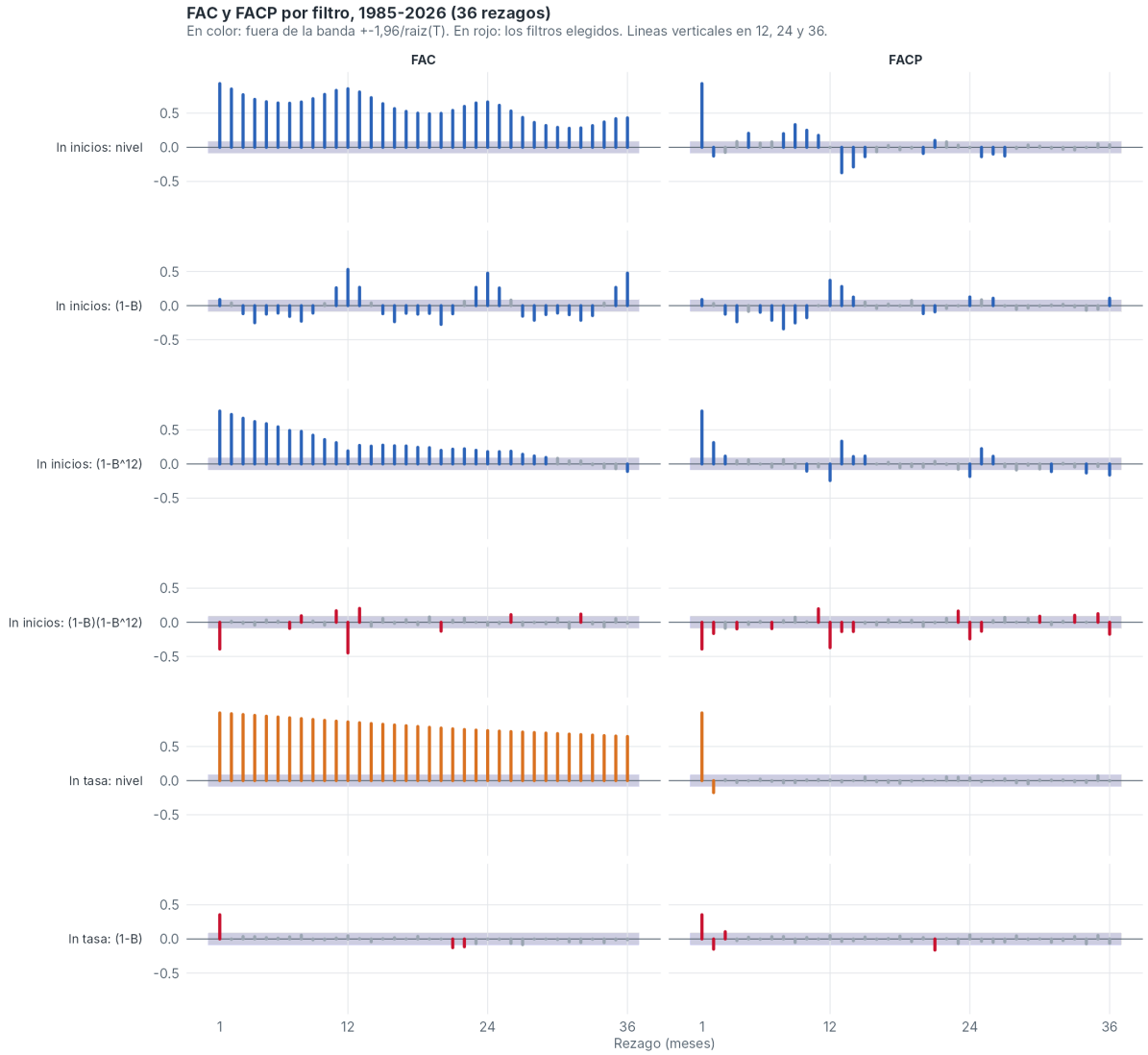


Figura 3: FAC y FACP por filtro (banda $\pm 1{,}96/\sqrt{T}$).

Filtro	FAC	Conclusión
x	$\rho_1 = 0{,}93$, $\rho_{24} = 0{,}66$	raíz regular y estacional
$(1-B)x$	$\rho_{12} = 0{,}53$, $\rho_{24} = 0{,}48$, $\rho_{36} = 0{,}48$	falta $(1-B^{12})$
$(1-B^{12})x$	gruesa, $\rho_1 = 0{,}77$	falta $(1-B)$ (regla 6)
$(1-B)(1-B^{12})x$	$\rho_1 = -0{,}39$, $\rho_{12} = -0{,}45$, resto ≈ 0	estacionaria; MA(1)×SMA(1)
y	$\rho_1 = 0{,}99$, decae lineal	raíz unitaria
$(1-B)y$	$\rho_1 = 0{,}35$, resto ≈ 0	estacionaria; MA(1)

Correlograma de LHOUST (nivel)

Sample: 1985M01 2026M08 Included observations: 500

Autocorrelation	Partial Correlation	k	AC	PAC	Q-Stat	Prob
		1	0.933	0.933	437.983	0.000
		2	0.854	-0.128	805.768	0.000
		3	0.770	-0.075	1105.398	0.000
		4	0.702	0.084	1354.915	0.000
		5	0.668	0.205	1580.873	0.000
		6	0.649	0.061	1795.196	0.000
		7	0.646	0.080	2007.770	0.000
		8	0.664	0.200	2232.667	0.000
		9	0.712	0.332	2491.959	0.000
		10	0.775	0.250	2799.693	0.000
		11	0.834	0.175	3156.505	0.000
		12	0.857	-0.002	3534.675	0.000
		13	0.810	-0.373	3873.031	0.000
		14	0.727	-0.288	4146.053	0.000
		15	0.639	-0.137	4357.306	0.000
		16	0.567	-0.065	4523.841	0.000
		17	0.526	0.026	4667.566	0.000
		18	0.500	-0.036	4797.829	0.000
		19	0.491	-0.010	4923.666	0.000
		20	0.497	-0.090	5052.968	0.000
		21	0.540	0.101	5205.857	0.000
		22	0.599	0.080	5394.312	0.000
		23	0.650	0.031	5616.311	0.000
		24	0.664	-0.005	5848.943	0.000

Correlograma de D(LHOUST,1,12)

Sample: 1986M02 2026M08 Included observations: 487

Autocorrelation	Partial Correlation	k	AC	PAC	Q-Stat	Prob
		1	-0.391	-0.391	75.103	0.000
		2	0.015	-0.164	75.210	0.000
		3	-0.014	-0.087	75.312	0.000
		4	-0.041	-0.096	76.138	0.000
		5	0.034	-0.030	76.710	0.000
		6	0.016	0.014	76.843	0.000
		7	-0.089	-0.093	80.801	0.000
		8	0.096	0.025	85.348	0.000
		9	0.019	0.074	85.526	0.000
		10	-0.042	0.007	86.387	0.000
		11	0.169	0.198	100.666	0.000
		12	-0.449	-0.371	201.651	0.000
		13	0.203	-0.136	222.425	0.000
		14	-0.057	-0.135	224.039	0.000
		15	0.055	-0.014	225.576	0.000
		16	-0.015	-0.038	225.685	0.000
		17	0.038	0.038	226.407	0.000
		18	-0.034	0.021	227.011	0.000
		19	0.078	0.029	230.107	0.000
		20	-0.130	-0.059	238.751	0.000
		21	0.030	-0.007	239.212	0.000
		22	0.059	0.057	240.997	0.000
		23	-0.001	0.165	240.997	0.000
		24	-0.043	-0.243	241.958	0.000

Figura 4: Correlogramas con Q de Ljung–Box: inicios en nivel y con $(1-B)(1-B^{12})$.

3.2. Dickey–Fuller aumentado

Esquema descendente (Dickey–Pantula): primero Δ, después el nivel (τ_τ, y τ_μ si la tendencia no es significativa). 12 rezagos; sensibilidad con 24 y con SBC. p-valores de MacKinnon.

Serie	Contraste	n	τ (12 rez.)	VC 5 %	p	Decisión	τ (24 rez.)	τ (SBC, p)
ln inicios	1: Δx (2da raíz)	486	−5,30	−2,87	$<$ 0,001	Rechaza	−3,37	−30,88 (0)
ln inicios	2: nivel, τ_τ	487	−1,79	−3,42	0,707	No rechaza	−2,53	−2,30 (1)
ln inicios	2: nivel, τ_μ	487	−1,86	−2,87	0,352	No rechaza	−2,52	−2,26 (1)
ln inicios	$(1-B^{12})x$	475	−3,65	−2,87	0,005	Rechaza	−2,90	−2,83 (14)
ln tasa	1: Δy (2da raíz)	486	−5,66	−2,87	$<$ 0,001	Rechaza	−4,89	−14,67 (1)
ln tasa	2: nivel, τ_τ	487	−1,96	−3,42	0,619	No rechaza	−1,35	−1,80 (3)
ln tasa	2: nivel, τ_μ	487	−1,98	−2,87	0,298	No rechaza	−1,65	−2,17 (2)

Cuadro 4: ADF, 1985–2026. Decisión con 12 rezagos, al 5 %.

- **Ambas series $I(1)$:** se rechaza sobre la diferencia (−5,30, −5,66) y no sobre el nivel (τ_μ = −1,86 y −1,98; VC −2,87).
- **El ADF no detecta la raíz estacional** (solo testea $z = 1$, *(Nota 6, Sec. 2.3)*): la decide el correlograma.
- **Sobre $(1-B^{12})x$ el ADF es frágil:** $p = 0{,}005$ con 12 rezagos, $p = 0{,}055$ con SBC. Se aplica $(1-B)$: manda el correlograma (regla 6).

Test ADF: LHOUST en nivel (tau_tau)

Null Hypothesis: LHOUST has a unit root
Exogenous: Constant, Linear Trend, Seasonal dummies (Jan-Nov)
Lag Length: 12 (fixed)
Dependent Variable: D(LHOUST)
Method: Least Squares
Sample: 1986M02 2026M08
Included observations: 487

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	**-1.792553**	**0.7071**
Test critical values:		
1% level	-3.976898	
5% level	-3.419119	
10% level	-3.132096	

*MacKinnon (1996) one-sided p-values.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	10.626700	6.239518	1.703128	0.0892
LHOUST(-1)	**-0.023103**	**0.012888**	**-1.792553**	**0.0737**
D(LHOUST(-1))	-0.353739	0.047093	-7.511425	0.0000
D(LHOUST(-2))	-0.132135	0.049917	-2.647074	0.0084
D(LHOUST(-3))	-0.079118	0.050259	-1.574202	0.1161
D(LHOUST(-4))	-0.087880	0.050584	-1.737330	0.0830
D(LHOUST(-5))	-0.019110	0.050844	-0.375861	0.7072
D(LHOUST(-6))	-0.000406	0.050768	-0.007990	0.9936
D(LHOUST(-7))	-0.041847	0.050760	-0.824421	0.4101
D(LHOUST(-8))	-0.018794	0.050770	-0.370179	0.7114
D(LHOUST(-9))	-0.002960	0.050617	-0.058477	0.9534
D(LHOUST(-10))	-0.009039	0.050305	-0.179678	0.8575
D(LHOUST(-11))	0.039581	0.049481	0.799916	0.4242
D(LHOUST(-12))	0.123717	0.046226	2.676370	0.0077
@TREND	0.000188	0.002836	0.066354	0.9471
Jan	-8.751493	2.175662	-4.022450	0.0001
Feb	-2.002818	2.207744	-0.907179	0.3648
Mar	15.605428	2.136764	7.303298	0.0000
Apr	14.175083	2.019004	7.020829	0.0000
May	8.021057	2.095433	3.827876	0.0001
Jun	5.865247	2.126965	2.757566	0.0061
Jul	0.588199	2.146152	0.274072	0.7842
Aug	-2.380490	2.136707	-1.114093	0.2658
Sep	-3.153801	2.147301	-1.468728	0.1426
Oct	0.336807	2.136212	0.157665	0.8748
Nov	-12.581955	2.115920	-5.946329	0.0000

R-squared	0.565243	Mean dependent var	-0.009265
Adjusted R-squared	0.541666	S.D. dependent var	12.601117
S.E. of regression	8.530998	Akaike info criterion	7.177200
Sum squared resid	33550.627052	Schwarz criterion	7.400803
Log likelihood	-1721.648189	Hannan-Quinn criter.	7.265040
F-statistic	23.974527	Durbin-Watson stat	2.032986
Prob(F-statistic)	0.000000		

Figura 5: ADF del nivel de los inicios. Resto en el Apéndice A.

3.3. Elección del filtro por SBC

Mismas fechas (desde 1986-02); los SBC son comparables porque todo $F(B)$ con $F(0) = 1$ tiene jacobiano unitario.

	Filtro	Mejor modelo admisible	$\hat{\sigma}$ (%)	Ljung–Box p (12/24/36)	SBC	ΔSBC
F1	$(1-B)x$ + dummies	ARMA$(0,1)(1,1)_{12}$	8,10	0,61 / 0,39 / 0,43	3526,0	49,1
F2	$(1-B^{12})x$	ARMA$(1,1)(0,1)_{12}$	8,22	0,26 / 0,30 / 0,35	3486,1	9,2
F2t	$(1-B^{12})x$ + tendencia	ARMA$(1,1)(0,1)_{12}$	8,22	0,26 / 0,31 / 0,36	3492,2	15,3
F3	$(1-B)(1-B^{12})x$	ARMA$(0,1)(0,1)_{12}$	8,25	0,41 / 0,47 / 0,53	3476,9	0,0

Cuadro 5: Mejor modelo admisible por filtro.

- **Gana** $(1-B)(1-B^{12})$: 9,2 puntos sobre F2 y 49,1 sobre dummies fijas.
- F2 necesita un AR(1) con raíz 1,012: la raíz unitaria que falta.

4. Inicios: ARIMA$(0,1,1)(0,1,1)_{12}$

4.1. Identificación

Rejilla de 48 modelos ARMA$(p \le 3, q \le 2)(P \le 1, Q \le 1)_{12}$ sobre $W_t = (1-B)(1-B^{12})x_t$; 20 admisibles. Gana por SBC el que sugiere el correlograma.

Modelo	k	$\hat{\sigma}$	LB12	LB24	LB36	Estac./invert.	AIC	SC	ΔSBC
ARMA(0,1)$(0,1)_{12}$	2	8,25	0,41	0,47	0,53	sí	7,1010	7,1268	0,0
ARMA(2,0)$(0,1)_{12}$	3	8,23	0,63	0,63	0,70	sí	7,0993	7,1337	3,4
ARMA(0,2)$(0,1)_{12}$	3	8,24	0,46	0,54	0,61	sí	7,1015	7,1359	4,4
ARMA(1,1)$(0,1)_{12}$	3	8,24	0,42	0,51	0,57	sí	7,1022	7,1366	4,8
ARMA(0,1)$(1,1)_{12}$	3	8,24	0,39	0,46	0,53	sí	7,1036	7,1380	5,5
ARMA(2,0)$(1,1)_{12}$	4	8,23	0,55	0,58	0,66	sí	7,1027	7,1457	9,2
ARMA(2,1)$(0,1)_{12}$	4	8,23	0,53	0,58	0,65	sí	7,1032	7,1462	9,5
ARMA(3,0)$(0,1)_{12}$	4	8,23	0,54	0,58	0,65	sí	7,1032	7,1462	9,5

Cuadro 6: Los ocho modelos de menor SBC.

Rezago	FAC empírica	FAC teórica	Fórmula (Nota 4)
1	−0,39	−0,37	$-\theta/(1+\theta^2)$
2	0,01	0,00	0
11	0,17	0,18	$\theta\Theta/[(1+\theta^2)(1+\Theta^2)]$
12	−0,45	−0,49	$-\Theta/(1+\Theta^2)$
13	0,20	0,18	$\theta\Theta/[(1+\theta^2)(1+\Theta^2)]$
24	−0,04	0,00	0

Cuadro 7: FAC empírica y teórica del modelo estimado.

4.2. Estimación

$$(1-B)(1-B^{12})\,x_t = \omega\,(1-B)(1-B^{12})\,I_t^{2020.04} + (1-\theta B)(1-\Theta B^{12})\,a_t$$

- $\hat{\theta} = 0{,}449$ (ee 0,039), $\hat{\Theta} = 0{,}838$ (ee 0,028), $\hat{\sigma}_a = 8{,}25\,\%$. Signo de las Notas; la salida usa $+\theta$.
- AO: $\hat{\omega} = -31{,}8$ (ee 6,8), es decir, $-27{,}3\,\%$.
- Invertible: raíz MA 2,23; raíces estacionales de módulo 1,015.

Inicios: ARIMA(0,1,1)(0,1,1)12 con AO 2020M04

Dependent Variable: D(LHOUST,1,12)
Method: ARMA Maximum Likelihood (exact, BFGS)
Sample: 1986M02 2026M08
Included observations: 487
Convergence achieved
Coefficient covariance computed using the observed Hessian

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	**-0.448734**	**0.038520**	**-11.649412**	**0.0000**
SMA(12)	**-0.838448**	**0.028190**	**-29.742607**	**0.0000**
AO2020M04	**-31.824774**	**6.840940**	**-4.652106**	**0.0000**

R-squared	0.528887	Mean dependent var	-0.022103
Adjusted R-squared	0.526941	S.D. dependent var	12.031865
S.E. of regression	8.275434	Akaike info criterion	7.100977
Sum squared resid	33145.680471	**Schwarz criterion**	**7.126777**
Log likelihood	-1726.087865	Hannan-Quinn criter.	7.111112
Jarque-Bera (Prob.)	4.027 (0.1335)	Durbin-Watson stat	2.051003

Inverted AR Roots	(none)
Inverted MA Roots	-0.49+/-0.85i 0.00+/-0.99i -0.85+/-0.49i 0.49+/-0.85i
	-0.99 0.85+/-0.49i 0.99 0.45

Figura 6: Modelo de los inicios.

4.3. Diagnóstico

- **Ljung–Box** ($K - 2$ gl): $p = 0{,}41$ / $0{,}47$ / $0{,}53$. Ruido blanco.
- **Jarque–Bera**: $4{,}0$ ($p = 0{,}133$). Sin el AO: 40, y 2020-04 es un residuo de $-5{,}1$ desvíos.
- **Sobreajuste**: nada significativo ($|t| \leq 1{,}32$); todo empeora el SBC.
- **Cuadrados**: $p < 0{,}001$. Varianza no constante (limitación).

Correlograma de los residuos: inicios

Sample: 1986M02 2026M08 Included observations: 487 Q-statistic probabilities adjusted for 2 ARMA term(s)

Autocorrelation	Partial Correlation	k	AC	PAC	Q-Stat	Prob
		1	-0.026	-0.026	0.330	
		2	0.024	0.023	0.613	
		3	0.066	0.067	2.726	0.099
		4	-0.002	0.001	2.728	0.256
		5	0.011	0.008	2.789	0.425
		6	0.016	0.012	2.916	0.572
		7	-0.042	-0.042	3.793	0.580
		8	0.071	0.068	6.317	0.389
		9	0.068	0.073	8.630	0.280
		10	-0.021	-0.016	8.855	0.355
		11	0.049	0.036	10.073	0.345
		12	0.026	0.021	10.399	0.406
		13	0.069	0.071	12.762	0.309
		14	0.027	0.020	13.116	0.361
		15	0.060	0.061	14.910	0.313
		16	0.011	0.006	14.975	0.380
		17	0.042	0.026	15.875	0.390
		18	0.003	-0.002	15.879	0.461
		19	0.028	0.024	16.266	0.505
		20	-0.064	-0.073	18.352	0.433
		21	0.025	0.011	18.670	0.478
		22	0.056	0.050	20.286	0.440
		23	0.052	0.055	21.666	0.419
		24	-0.019	-0.034	21.858	0.468
		25	0.048	0.031	23.029	0.459
		26	0.116	0.105	29.924	0.187
		27	-0.036	-0.043	30.580	0.203
		28	0.012	-0.006	30.652	0.241
		29	-0.005	-0.008	30.665	0.285
		30	0.015	0.003	30.783	0.327
		31	0.001	-0.015	30.784	0.376
		32	0.013	0.010	30.871	0.422
		33	-0.021	-0.008	31.104	0.461
		34	-0.025	-0.063	31.426	0.495
		35	0.025	0.016	31.746	0.529
		36	-0.045	-0.045	32.813	0.526

Figura 7: Correlograma de los residuos (ajustado por 2 términos ARMA).

Parámetro agregado	Estimación	t	ΔSBC
AR(1)	−0,113	−1,22	4,8
MA(2)	0,064	1,32	4,4
SAR(12)	0,045	0,84	5,5
SMA(24)	−0,039	−0,85	5,5

Cuadro 8: Sobreajuste *(Nota 6, Sec. 10.4)*.

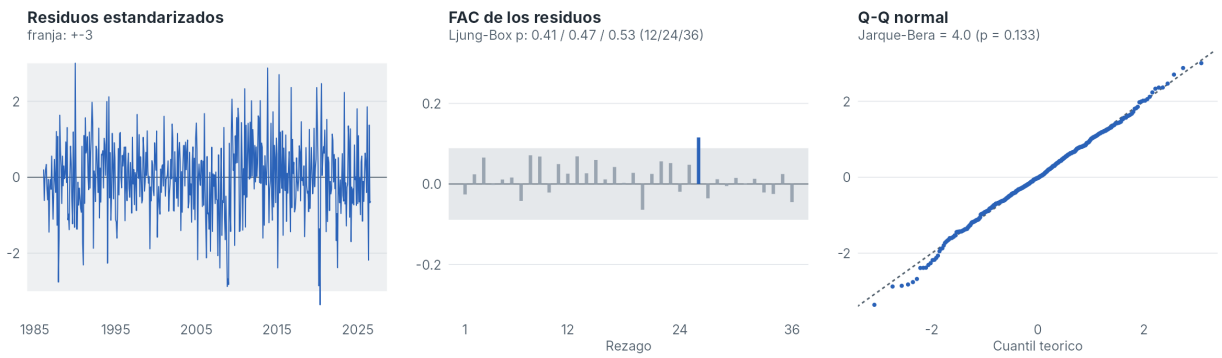


Figura 8: Residuos estandarizados, FAC y Q–Q.

4.4. Estabilidad y serie desestacionalizada

Muestra	$\hat{\theta}$ (ee)	$\hat{\Theta}$ (ee)	$\hat{\sigma}$	n
1986–2026 (modelo elegido)	0,45 (0,04)	0,84 (0,03)	8,25	487
1986–2007	0,49 (0,05)	0,80 (0,04)	7,46	263
2008–2026	0,42 (0,06)	0,81 (0,06)	9,27	224

Cuadro 9: El modelo por subperíodos (regla 9).

- **Dinámica estable:** $\hat{\theta}$ 0,49 → 0,42, $\hat{\Theta}$ 0,80 → 0,81. Sube $\hat{\sigma}$: 7,46 → 9,27.

- **Desestacionalizada (regla 8):** conserva $\rho_{12} = -0{,}16$; su mejor modelo, ARMA(1,2)(1,1)$_{12}$, necesita parte estacional. Se descarta (Apéndice A).

5. Tasa: ARIMA(0,1,1)

Correlograma de LMORT (nivel)

Sample: 1985M01 2026M08 Included observations: 500

Autocorrelation	Partial Correlation	k	AC	PAC	Q-Stat	Prob
		1	0.991	0.991	494.313	0.000
		2	0.980	-0.176	978.059	0.000
		3	0.968	0.013	1451.157	0.000
		4	0.956	-0.028	1913.377	0.000
		5	0.944	-0.002	2364.786	0.000
		6	0.932	0.022	2805.945	0.000
		7	0.920	-0.011	3237.027	0.000
		8	0.908	-0.032	3657.726	0.000
		9	0.896	-0.029	4067.678	0.000
		10	0.883	0.010	4467.126	0.000
		11	0.871	0.016	4856.564	0.000
		12	0.859	0.009	5236.425	0.000
		13	0.848	-0.019	5606.745	0.000
		14	0.836	0.001	5967.718	0.000
		15	0.825	0.053	6320.318	0.000
		16	0.815	-0.016	6664.724	0.000
		17	0.804	-0.026	7000.702	0.000
		18	0.793	-0.039	7327.846	0.000
		19	0.781	-0.008	7646.111	0.000
		20	0.770	0.018	7955.842	0.000
		21	0.759	0.005	8257.346	0.000
		22	0.749	0.054	8551.602	0.000
		23	0.740	0.050	8839.760	0.000
		24	0.733	0.039	9122.714	0.000

Correlograma de D(LMORT)

Sample: 1985M02 2026M08 Included observations: 499

Autocorrelation	Partial Correlation	k	AC	PAC	Q-Stat	Prob
		1	0.355	0.355	63.153	0.000
		2	-0.005	-0.150	63.166	0.000
		3	0.036	0.106	63.806	0.000
		4	0.033	-0.024	64.364	0.000
		5	0.019	0.025	64.546	0.000
		6	0.012	-0.004	64.613	0.000
		7	0.031	0.033	65.090	0.000
		8	0.054	0.035	66.597	0.000
		9	-0.012	-0.051	66.671	0.000
		10	-0.012	0.022	66.747	0.000
		11	0.013	0.000	66.838	0.000
		12	0.046	0.049	67.943	0.000
		13	0.005	-0.036	67.955	0.000
		14	-0.041	-0.026	68.811	0.000
		15	0.004	0.028	68.820	0.000
		16	0.017	-0.005	68.978	0.000
		17	0.001	0.006	68.979	0.000
		18	0.037	0.041	69.698	0.000
		19	0.001	-0.038	69.699	0.000
		20	-0.001	0.019	69.700	0.000
		21	-0.127	-0.163	78.199	0.000
		22	-0.115	0.005	85.076	0.000
		23	-0.068	-0.068	87.515	0.000
		24	-0.001	0.058	87.515	0.000

Figura 9: Correlogramas de la tasa: nivel y primera diferencia.

Modelo	Const.	$\hat{\sigma}$	LB12	LB24	LB36	Estac./invert.	AIC	SC	ΔSBC
ARMA(0,1)	no	3,13	0,93	0,32	0,37	sí	5,1262	5,1346	0,0
ARMA(0,1)	sí	3,13	0,93	0,32	0,37	sí	5,1293	5,1462	5,8
ARMA(1,1)	no	3,13	0,91	0,24	0,31	sí	5,1296	5,1465	5,9
ARMA(0,2)	no	3,13	0,91	0,25	0,32	sí	5,1296	5,1465	5,9
ARMA(2,0)	no	3,15	0,44	0,20	0,26	sí	5,1410	5,1578	11,6
ARMA(1,1)	sí	3,13	0,91	0,24	0,32	sí	5,1326	5,1580	11,7
ARMA(0,2)	sí	3,13	0,91	0,25	0,32	sí	5,1327	5,1580	11,7
ARMA(3,0)	no	3,13	0,88	0,24	0,32	sí	5,1334	5,1587	12,0

Cuadro 10: Los ocho modelos de menor SBC sobre $(1-B)y_t$.

$$(1-B)\,y_t = (1-\theta B)\,a_t, \qquad \hat{\theta} = -0{,}439\ (0{,}041), \qquad \hat{\sigma}_a = 3{,}13\,\%$$

- Sin constante (empeora el SBC). $\hat{\theta} < 0 \Rightarrow \rho_1 = 0{,}37$: inercia de un mes.
- **Ljung–Box**: $p = 0{,}93$ / $0{,}32$ / $0{,}37$. Sobreajuste y subperíodos confirman el MA(1).
- **Limitación:** colas pesadas (curtosis 6,0, JB = 246) y varianza cambiante (cuadrados: $p < 0{,}001$). Residuos extremos: $2008-12(-3{,}9)$; $2013-06(3{,}9)$; $2022-04(4{,}5)$; $2022-09(5{,}2)$ (Fed compra MBS, *taper tantrum*, suba de tasas 2022).

Tasa: ARIMA(0,1,1)

Dependent Variable: D(LMORT)
Method: ARMA Maximum Likelihood (exact, BFGS)
Sample: 1985M02 2026M08
Included observations: 499
Convergence achieved
Coefficient covariance computed using the observed Hessian

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	**0.439069**	**0.041310**	**10.628757**	**0.0000**

R-squared	0.153935	Mean dependent var	-0.134961
Adjusted R-squared	0.153935	S.D. dependent var	3.409301
S.E. of regression	3.135935	Akaike info criterion	5.126163
Sum squared resid	4897.376129	**Schwarz criterion**	**5.134605**
Log likelihood	-1277.977675	Hannan-Quinn criter.	5.129476
Jarque-Bera (Prob.)	246.157 (0.0000)	Durbin-Watson stat	2.019702

Inverted AR Roots	(none)
Inverted MA Roots	-0.44

Figura 10: Modelo de la tasa.

Correlograma de los residuos: tasa

Sample: 1985M02 2026M08 Included observations: 499 Q-statistic probabilities adjusted for 1 ARMA term(s)

Autocorrelation	Partial Correlation	k	AC	PAC	Q-Stat	Prob
		1	-0.011	-0.011	0.062	
		2	-0.013	-0.013	0.152	0.697
		3	0.034	0.034	0.744	0.689
		4	0.017	0.017	0.885	0.829
		5	0.011	0.012	0.941	0.919
		6	0.007	0.006	0.963	0.966
		7	0.003	0.003	0.969	0.987
		8	0.066	0.066	3.218	0.864
		9	-0.038	-0.037	3.947	0.862
		10	0.004	0.004	3.954	0.914
		11	-0.005	-0.011	3.967	0.949
		12	**0.046**	**0.046**	**5.035**	**0.929**
		13	0.005	0.005	5.049	0.956
		14	-0.047	-0.046	6.185	0.939
		15	0.012	0.009	6.261	0.959
		16	0.025	0.018	6.583	0.968
		17	-0.032	-0.024	7.115	0.971
		18	0.066	0.065	9.388	0.927
		19	-0.048	-0.049	10.606	0.910
		20	0.067	0.065	12.970	0.840
		21	-0.136	-0.142	22.696	0.304
		22	-0.041	-0.031	23.564	0.315
		23	-0.060	-0.077	25.442	0.276
		24	**0.021**	**0.026**	**25.673**	**0.317**
		25	-0.002	0.005	25.675	0.370
		26	-0.032	-0.030	26.228	0.396
		27	-0.084	-0.074	29.985	0.268
		28	0.037	0.022	30.696	0.284
		29	-0.016	0.009	30.829	0.325
		30	0.006	0.007	30.847	0.373
		31	-0.021	-0.017	31.088	0.411
		32	-0.060	-0.067	32.997	0.370
		33	0.044	0.053	34.021	0.371
		34	-0.072	-0.066	36.830	0.296
		35	0.007	0.024	36.855	0.338
		36	**0.022**	**-0.006**	**37.117**	**0.372**

Figura 11: Correlograma de los residuos de la tasa (ajustado por 1 término ARMA).

Parámetro agregado	Estimación	t	ΔSBC
AR(1)	$-0{,}055$	$-0{,}55$	5,9
MA(2)	$-0{,}023$	$-0{,}53$	5,9

Cuadro 11: Sobreajuste.

Muestra	$\hat{\theta}$ (ee)	$\hat{\sigma}$	n
1985–2026 (modelo elegido)	$-0{,}44$ (0,04)	3,13	499
1985–2007	$-0{,}46$ (0,05)	2,56	275
2008–2026	$-0{,}43$ (0,06)	3,72	224

Cuadro 12: Subperíodos.

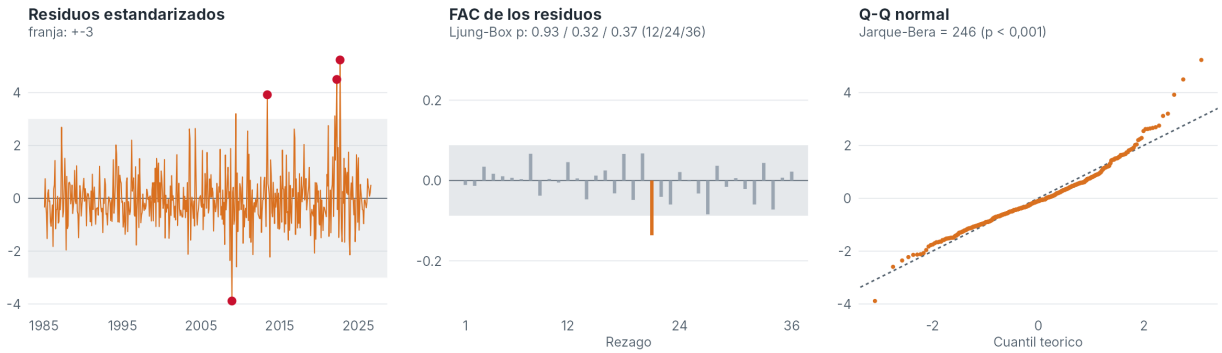


Figura 12: Residuos de la tasa.

6. Pronóstico fuera de muestra

Estimación 1985–2017, evaluación 2018–2019 (24 orígenes móviles, $h = 1, 3, 6, 12$). Orden elegido solo con la muestra de estimación (coincide con el final).

Modelo	RMSE h=1	h=3	h=6	h=12	MAPE h=1	h=3	h=6	h=12	U (h=1)
Líneas aéreas (F3)	7,1	7,4	9,6	11,7	5,1	5,4	7,8	8,6	0,034
Rival F2	7,1	7,5	9,5	11,6	5,0	5,4	7,4	7,6	0,034
Promedio F3 y F2	7,1	7,4	9,5	11,5	5,1	5,4	7,6	8,0	0,034
Ingenuo estacional	10,5	10,7	10,4	11,9	7,5	7,6	7,0	7,9	0,049
Paseo aleatorio estacional	8,7	11,2	13,9	16,9	6,6	8,4	11,4	12,8	0,043

Cuadro 13: Inicios: RMSE (%) y MAPE (%, sobre el nivel); U de Theil a un paso.

Modelo	RMSE h=1	h=3	h=6	h=12	MAPE h=1	h=3	h=6	h=12
ARIMA(0,1,1)	2,72	6,82	11,49	18,37	2,07	5,88	10,16	17,90
Paseo aleatorio	3,04	7,20	11,88	18,35	2,36	6,20	10,66	17,85

Cuadro 14: Tasa: RMSE (%) y MAPE (%).

- **Inicios:** mejora sobre el paseo aleatorio estacional de 19, 34, 31 y 31 %. A 12 meses empata con el ingenuo estacional. Combinar con F2 no agrega.
- **Tasa:** mejora 10 % a un mes y nada a 12 (pronóstico plano tras el primer paso, *(Nota 6, Sec. 3.4)*).

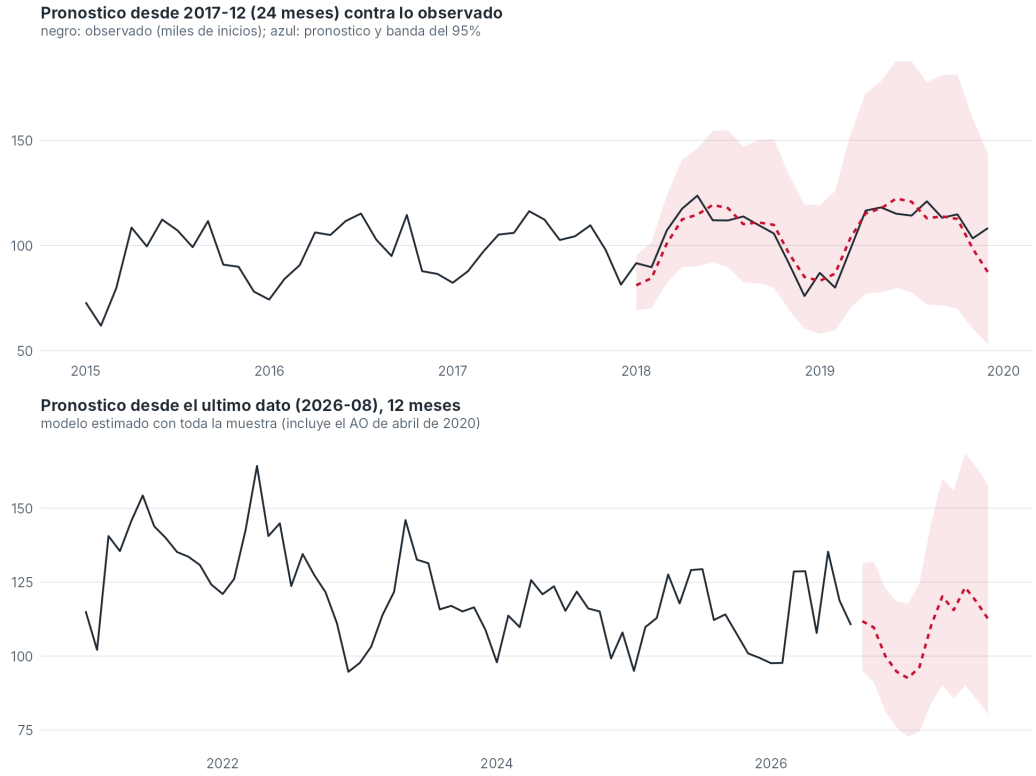


Figura 13: Arriba: desde 2017-12 contra lo observado (100 % dentro de la banda). Abajo: 12 meses desde 2026-08.

7. Relación entre las series

7.1. Niveles: regresión espuria

Regresión de ln inicios sobre ln tasa	Pendiente	t	R^2	d_W	¿$R^2 > d_W$?
Niveles	0,312	7,71	0,107	0,149	No
Niveles + dummies mensuales	0,305	8,15	0,255	0,092	Sí
$(1-B)(1-B^{12})$ en ambas (+ AO)	−0,126	−1,10	0,019	2,819	No

Cuadro 15: Regresiones de ln inicios sobre ln tasa.

- **Niveles:** pendiente positiva ($t = 7{,}7$), signo equivocado; $d_W = 0{,}149$. Con dummies, $R^2 = 0{,}255 > d_W = 0{,}092$.
- **Filtradas:** la relación contemporánea desaparece ($R^2 = 0{,}019$).

Regresion en niveles: LHOUST sobre LMORT

Dependent Variable: LHOUST
Method: Least Squares
Sample: 1985M01 2026M08
Included observations: 500

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	410.731601	7.514932	54.655398	0.0000
LMORT	**0.312325**	**0.040484**	**7.714771**	**0.0000**

R-squared	**0.106755**	Mean dependent var	467.582996
Adjusted R-squared	0.104961	S.D. dependent var	34.814432
S.E. of regression	32.936714	Akaike info criterion	9.831045
Sum squared resid	540243.902388	Schwarz criterion	9.847903
Log likelihood	-2455.761239	Hannan-Quinn criter.	9.837660
F-statistic	59.517686	**Durbin-Watson stat**	**0.148549**
Prob(F-statistic)	0.000000		

Figura 14: Regresión en niveles.

7.2. Innovaciones: correlación cruzada y Haugh

Residuos de los dos modelos univariados; banda ±0,089; $S = n\sum_{-M}^{M} r_k^2 \sim \chi^2_{2M+1}$.

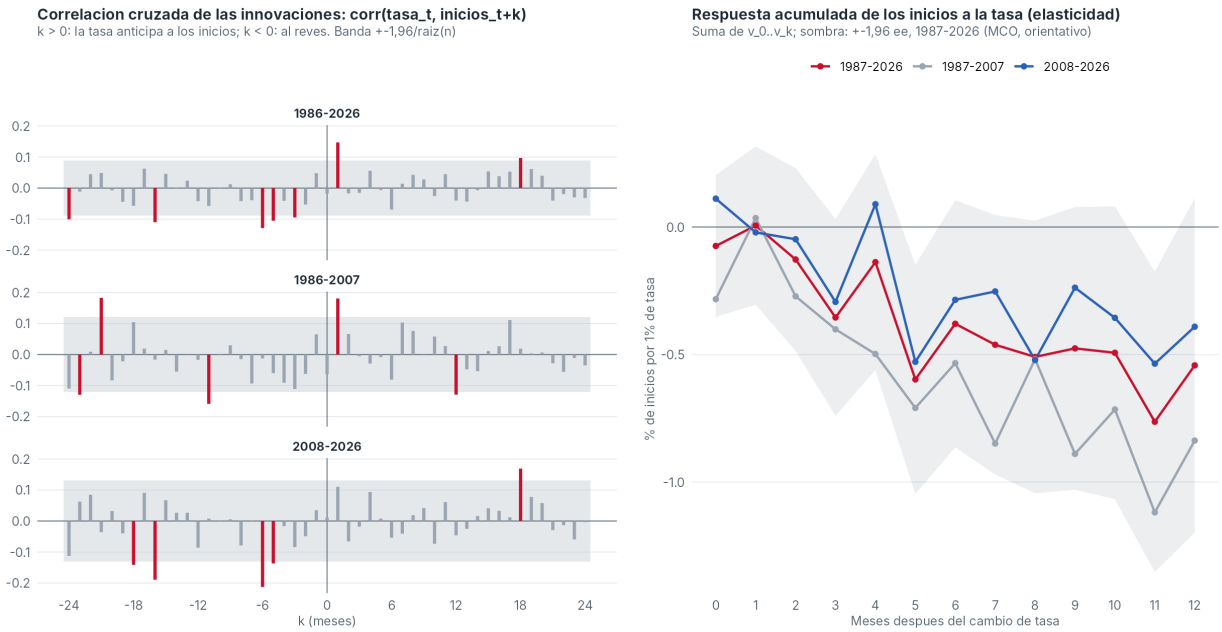


Figura 15: Izquierda: correlación cruzada de las innovaciones. Derecha: respuesta acumulada.

Período	n	S, M=6 (gl)	p	S, M=12 (gl)	p	p, solo $k \geq 0$	p, solo $k < 0$
1986–2026	487	36,0 (13)	$< 0{,}001$	43,6 (25)	0,012	0,147	0,014
1986–2007	263	21,3 (13)	0,067	40,4 (25)	0,027	0,047	0,121
2008–2026	224	23,2 (13)	0,040	29,6 (25)	0,239	0,714	0,070

Cuadro 16: Test de Haugh. Las dos últimas columnas separan la dirección ($M = 12$).

- **No son independientes** ($M = 6$: $p < 0{,}001$).
- **Retroalimentación:** en toda la muestra domina inicios → tasa ($k < 0$: $p = 0{,}014$; $k \geq 0$: $p = 0{,}147$). Tasa → inicios solo en 1986–2007 ($p = 0{,}047$).

7.3. Rezagos distribuidos con el mismo filtro

$F(B) = (1-B)(1-B^{12})$ en ambas series: $F(B)x_t = v(B)F(B)y_t + F(B)n_t$, con $v(B)$ intacto. MCO con rezagos 0–12 y el AO.

Rezagos distribuidos 0-12, filtro (1-B)(1-B^12) en ambas

Dependent Variable: D(LHOUST,1,12)
Method: Least Squares
Sample: 1987M02 2026M08
Included observations: 475

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.035179	0.547363	0.064270	0.9488
D(LMORT,1,12)	-0.074394	0.141831	-0.524525	0.6002
D(LMORT(-1),1,12)	0.080220	0.137663	0.582729	0.5604
D(LMORT(-2),1,12)	-0.133172	0.138808	-0.959397	0.3379
D(LMORT(-3),1,12)	-0.227296	0.139917	-1.624509	0.1050
D(LMORT(-4),1,12)	0.216915	0.140456	1.544357	0.1232
D(LMORT(-5),1,12)	**-0.459884**	**0.140642**	**-3.269894**	**0.0012**
D(LMORT(-6),1,12)	0.218226	0.140654	1.551502	0.1215
D(LMORT(-7),1,12)	-0.082319	0.140467	-0.586038	0.5581
D(LMORT(-8),1,12)	-0.047676	0.139851	-0.340910	0.7333
D(LMORT(-9),1,12)	0.033469	0.140261	0.238623	0.8115
D(LMORT(-10),1,12)	-0.017243	0.139873	-0.123273	0.9019
D(LMORT(-11),1,12)	**-0.270509**	**0.137571**	**-1.966325**	**0.0499**
D(LMORT(-12),1,12)	0.221151	0.141303	1.565085	0.1183
D(AO2020M04,1,12)	**-18.396111**	**6.051037**	**-3.040158**	**0.0025**

R-squared	0.064701	Mean dependent var	0.016833
Adjusted R-squared	0.036236	S.D. dependent var	12.148751
S.E. of regression	11.926611	Akaike info criterion	7.826491
Sum squared resid	65432.261170	Schwarz criterion	7.957964
Log likelihood	-1843.791580	Hannan-Quinn criter.	7.878193
F-statistic	2.272963	Durbin-Watson stat	2.839457
Prob(F-statistic)	0.005253		

Figura 16: Rezagos distribuidos 0–12 (elasticidades).

Meses	Elasticidad acumulada (ee)	% de inicios por +1 pp	1987–2007	2008–2026
0	−0,07 (0,14)	−1,0	−0,28 (0,22)	0,11 (0,20)
3	−0,35 (0,20)	−5,0	−0,40 (0,29)	−0,29 (0,28)
6	−0,38 (0,25)	−5,3	−0,53 (0,38)	−0,29 (0,34)
12	−0,54 (0,33)	−7,6	−0,84 (0,53)	−0,39 (0,45)

Cuadro 17: Elasticidad acumulada. Tercera columna: efecto de +1 pp sobre 6,67 % (+14,0 % de la tasa).

- **Elasticidad:** −0,35 (ee 0,20) a 3 meses, −0,54 (ee 0,33) a 12.
- **En puntos:** +1 pp ⇒ −7,6 % de inicios a un año; IC 95 % [−17 %; 2 %].
- Errores estándar orientativos ($d_W = 2{,}84$): corresponde función de transferencia (segunda parte).

8. Limitaciones y pasos siguientes

- **Exogeneidad:** hay retroalimentación. Opciones: acotar a 1986–2007 o usar el bono a 10 años como causa.
- **Intervenciones:** 2020-04 (inicios); 2008-12, 2013-06 y 2022 (tasa).

- **Varianza:** heterocedasticidad en ambas series; colas pesadas en la tasa.
- **Estacionalidad:** la raíz estacional se decidió por correlograma y SBC (HEGY fuera del programa).

Fuentes: U.S. Census Bureau, Freddie Mac PMMS, FRED. Teoría: Notas 1 a 6 y apunte de la cátedra.

A. Salidas complementarias

Test ADF: D(LHOUST)

Null Hypothesis: D(LHOUST) has a unit root
Exogenous: Constant, Seasonal dummies (Jan-Nov)
Lag Length: 12 (fixed)
Dependent Variable: D(D(LHOUST))
Method: Least Squares
Sample: 1986M03 2026M08
Included observations: 486

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		**-5.298984**	**0.0000**
Test critical values:			
1% level		-3.443692	
5% level		-2.867280	
10% level		-2.569870	

*MacKinnon (1996) one-sided p-values.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.088378	0.384297	-0.229973	0.8182
D(LHOUST)(-1)	**-1.460525**	**0.275624**	**-5.298984**	**0.0000**
D(D(LHOUST)(-1))	0.073998	0.266142	0.278040	0.7811
D(D(LHOUST)(-2))	-0.075154	0.252976	-0.297082	0.7665
D(D(LHOUST)(-3))	-0.167528	0.238297	-0.703021	0.4824
D(D(LHOUST)(-4))	-0.262719	0.222693	-1.179733	0.2387
D(D(LHOUST)(-5))	-0.288860	0.206252	-1.400522	0.1620
D(D(LHOUST)(-6))	-0.292185	0.189000	-1.545954	0.1228
D(D(LHOUST)(-7))	-0.343199	0.171095	-2.005895	0.0455
D(D(LHOUST)(-8))	-0.370741	0.151247	-2.451230	0.0146
D(D(LHOUST)(-9))	-0.369805	0.128781	-2.871591	0.0043
D(D(LHOUST)(-10))	-0.368932	0.104781	-3.521002	0.0005
D(D(LHOUST)(-11))	-0.310298	0.078123	-3.971898	0.0001
D(D(LHOUST)(-12))	-0.140918	0.046102	-3.056620	0.0024
Jan	-7.485015	2.181306	-3.431437	0.0007
Feb	-2.071468	2.187665	-0.946885	0.3442
Mar	14.168534	2.185007	6.484435	0.0000
Apr	11.127157	2.238221	4.971430	0.0000
May	6.797434	2.107233	3.225764	0.0013
Jun	5.537645	2.109187	2.625488	0.0089
Jul	0.370779	2.125784	0.174420	0.8616
Aug	-1.929333	2.130222	-0.905696	0.3656
Sep	-2.609396	2.141043	-1.218750	0.2236
Oct	1.081640	2.134742	0.506684	0.6126
Nov	-11.703955	2.119436	-5.522203	0.0000

R-squared	0.763038	Mean dependent var	0.000620
Adjusted R-squared	0.750701	S.D. dependent var	16.954618
S.E. of regression	8.465407	Akaike info criterion	7.159923
Sum squared resid	33036.699855	Schwarz criterion	7.375263
Log likelihood	-1714.861396	Hannan-Quinn criter.	7.244525
F-statistic	61.852450	Durbin-Watson stat	2.019190
Prob(F-statistic)	0.000000		

Test ADF: D(LHOUST,0,12)

Null Hypothesis: D(LHOUST,0,12) has a unit root
Exogenous: Constant
Lag Length: 12 (fixed)
Dependent Variable: D(D(LHOUST,0,12))
Method: Least Squares
Sample: 1987M02 2026M08
Included observations: 475

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		**-3.654956**	**0.0051**
Test critical values:			
1% level		-3.444004	
5% level		-2.867418	
10% level		-2.569944	

*MacKinnon (1996) one-sided p-values.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.067220	0.451178	-0.148987	0.8816
D(LHOUST,0,12)(-1)	**-0.119686**	**0.032746**	**-3.654956**	**0.0003**
D(D(LHOUST,0,12)(-1))	-0.325176	0.047894	-6.789469	0.0000
D(D(LHOUST,0,12)(-2))	-0.129550	0.050185	-2.581429	0.0101
D(D(LHOUST,0,12)(-3))	-0.049118	0.050491	-0.972794	0.3312
D(D(LHOUST,0,12)(-4))	-0.004031	0.050450	-0.079893	0.9364
D(D(LHOUST,0,12)(-5))	0.016300	0.050431	0.323221	0.7467
D(D(LHOUST,0,12)(-6))	0.050581	0.050210	1.007383	0.3143
D(D(LHOUST,0,12)(-7))	0.004297	0.050137	0.085703	0.9317
D(D(LHOUST,0,12)(-8))	0.110020	0.049923	2.203774	0.0280
D(D(LHOUST,0,12)(-9))	0.119005	0.049870	2.386308	0.0174
D(D(LHOUST,0,12)(-10))	0.064970	0.049609	1.309631	0.1910
D(D(LHOUST,0,12)(-11))	0.072562	0.048320	1.501701	0.1339
D(D(LHOUST,0,12)(-12))	-0.341844	0.044031	-7.763775	0.0000

R-squared	0.364294	Mean dependent var	0.016833
Adjusted R-squared	0.346367	S.D. dependent var	12.148751
S.E. of regression	9.821969	Akaike info criterion	7.436151
Sum squared resid	44473.163584	Schwarz criterion	7.558859
Log likelihood	-1752.085827	Hannan-Quinn criter.	7.484406
F-statistic	20.321368	Durbin-Watson stat	2.069943
Prob(F-statistic)	0.000000		

Figura 17: ADF sobre Δx (paso 1 de Dickey–Pantula) y sobre $(1 - B^{12})x$.

Test ADF: LMORT en nivel (tau_tau)

Null Hypothesis: LMORT has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 12 (fixed)
Dependent Variable: D(LMORT)
Method: Least Squares
Sample: 1986M02 2026M08
Included observations: 487

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	**-1.963178**	**0.6194**
Test critical values:		
1% level	-3.976898	
5% level	-3.419119	
10% level	-3.132096	

*MacKinnon (1996) one-sided p-values.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	2.929439	1.689828	1.733572	0.0836
LMORT(-1)	**-0.013936**	**0.007099**	**-1.963178**	**0.0502**
D(LMORT(-1))	0.429235	0.045897	9.352036	0.0000
D(LMORT(-2))	-0.193174	0.049918	-3.869846	0.0001
D(LMORT(-3))	0.132425	0.050684	2.612747	0.0093
D(LMORT(-4))	-0.034473	0.050908	-0.677166	0.4986
D(LMORT(-5))	0.030669	0.050884	0.602721	0.5470
D(LMORT(-6))	0.000185	0.050898	0.003630	0.9971
D(LMORT(-7))	0.008814	0.050911	0.173124	0.8626
D(LMORT(-8))	0.070793	0.050824	1.392912	0.1643
D(LMORT(-9))	-0.058501	0.050869	-1.150033	0.2507
D(LMORT(-10))	0.039127	0.050711	0.771567	0.4408
D(LMORT(-11))	-0.015458	0.049714	-0.310949	0.7560
D(LMORT(-12))	0.059751	0.046199	1.293361	0.1965
@TREND	-0.001802	0.001797	-1.003101	0.3163

R-squared	0.169370	Mean dependent var	-0.100664
Adjusted R-squared	0.144733	S.D. dependent var	3.426785
S.E. of regression	3.169113	Akaike info criterion	5.175097
Sum squared resid	4740.426537	Schwarz criterion	5.304099
Log likelihood	-1245.136115	Hannan-Quinn criter.	5.225774
F-statistic	6.874542	Durbin-Watson stat	1.995883
Prob(F-statistic)	0.000000		

Test ADF: D(LMORT)

Null Hypothesis: D(LMORT) has a unit root
Exogenous: Constant
Lag Length: 12 (fixed)
Dependent Variable: D(D(LMORT))
Method: Least Squares
Sample: 1986M03 2026M08
Included observations: 486

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	**-5.661074**	**0.0000**
Test critical values:		
1% level	-3.443692	
5% level	-2.867280	
10% level	-2.569870	

*MacKinnon (1996) one-sided p-values.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.058119	0.145207	-0.400250	0.6892
D(LMORT)(-1)	**-0.633744**	**0.111948**	**-5.661074**	**0.0000**
D(D(LMORT)(-1))	0.060808	0.109821	0.553698	0.5800
D(D(LMORT)(-2))	-0.140988	0.105608	-1.335009	0.1825
D(D(LMORT)(-3))	-0.013354	0.101802	-0.131177	0.8957
D(D(LMORT)(-4))	-0.056420	0.097156	-0.580721	0.5617
D(D(LMORT)(-5))	-0.029890	0.093284	-0.320421	0.7488
D(D(LMORT)(-6))	-0.036029	0.088295	-0.408059	0.6834
D(D(LMORT)(-7))	-0.033981	0.082861	-0.410100	0.6819
D(D(LMORT)(-8))	0.030788	0.077063	0.399516	0.6897
D(D(LMORT)(-9))	-0.036055	0.070239	-0.513326	0.6080
D(D(LMORT)(-10))	-0.000090	0.063656	-0.001407	0.9989
D(D(LMORT)(-11))	-0.028492	0.052829	-0.539323	0.5899
D(D(LMORT)(-12))	0.035768	0.045860	0.779944	0.4358

R-squared	0.353296	Mean dependent var	0.007264
Adjusted R-squared	0.335484	S.D. dependent var	3.905227
S.E. of regression	3.183457	Akaike info criterion	5.182196
Sum squared resid	4783.435145	Schwarz criterion	5.302786
Log likelihood	-1245.273586	Hannan-Quinn criter.	5.229572
F-statistic	19.834968	Durbin-Watson stat	1.994970
Prob(F-statistic)	0.000000		

Figura 18: ADF de la tasa: nivel (τ_τ) y primera diferencia.

Correlograma de D(LHOUST)

Sample: 1985M02 2026M08 Included observations: 499

Autocorrelation	Partial Correlation	k	AC	PAC	Q-Stat	Prob
		1	0.089	0.089	4.020	0.045
		2	0.038	0.031	4.762	0.092
		3	-0.118	-0.125	11.748	0.008
		4	-0.251	-0.236	43.473	0.000
		5	-0.124	-0.084	51.260	0.000
		6	-0.109	-0.097	57.331	0.000
		7	-0.158	-0.214	69.981	0.000
		8	-0.228	-0.340	96.445	0.000
		9	-0.109	-0.253	102.534	0.000
		10	0.031	-0.177	103.013	0.000
		11	0.261	0.001	137.944	0.000
		12	**0.530**	**0.371**	**282.382**	**0.000**
		13	0.268	0.281	319.359	0.000
		14	0.038	0.126	320.112	0.000
		15	-0.118	0.051	327.359	0.000
		16	-0.235	-0.041	356.053	0.000
		17	-0.112	0.022	362.554	0.000
		18	-0.125	-0.005	370.640	0.000
		19	-0.114	0.075	377.389	0.000
		20	-0.275	-0.117	416.866	0.000
		21	-0.120	-0.091	424.335	0.000
		22	0.062	-0.041	426.357	0.000
		23	0.267	-0.003	463.928	0.000
		24	**0.475**	**0.127**	**582.850**	**0.000**

Correlograma de D(LHOUST_SA)

Sample: 1986M02 2026M08 Included observations: 487

Autocorrelation	Partial Correlation	k	AC	PAC	Q-Stat	Prob
		1	-0.350	-0.350	59.858	0.000
		2	0.004	-0.134	59.867	0.000
		3	0.003	-0.049	59.871	0.000
		4	-0.007	-0.026	59.894	0.000
		5	0.035	0.028	60.498	0.000
		6	0.019	0.049	60.673	0.000
		7	-0.053	-0.028	62.073	0.000
		8	0.069	0.050	64.449	0.000
		9	0.013	0.060	64.535	0.000
		10	-0.007	0.030	64.559	0.000
		11	0.074	0.099	67.324	0.000
		12	**-0.158**	**-0.109**	**79.884**	**0.000**
		13	0.127	0.041	87.934	0.000
		14	-0.031	0.006	88.409	0.000
		15	0.035	0.044	89.021	0.000
		16	0.036	0.070	89.675	0.000
		17	0.005	0.058	89.688	0.000
		18	-0.021	0.012	89.904	0.000
		19	0.089	0.082	93.940	0.000
		20	-0.094	-0.026	98.401	0.000
		21	0.018	-0.028	98.562	0.000
		22	0.068	0.057	100.922	0.000
		23	0.045	0.117	101.954	0.000
		24	**-0.130**	**-0.123**	**110.633**	**0.000**

Figura 19: Correlogramas de $(1 - B)x$ (picos estacionales que no decaen) y de la serie desestacionalizada en diferencia.

Inicios desestacionalizados: (1-B)x, ARMA(1,2)(1,1)_12

Dependent Variable: D(LHOUST_SA)
Method: ARMA Maximum Likelihood (exact, BFGS)
Sample: 1986M02 2026M08
Included observations: 487
Convergence achieved
Coefficient covariance computed using the observed Hessian

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.966283	0.019186	50.363055	0.0000
MA(1)	-1.402394	0.045590	-30.761249	0.0000
MA(2)	0.454168	0.041601	10.917175	0.0000
SAR(12)	**0.496744**	**0.114342**	**4.344376**	**0.0000**
SMA(12)	**-0.725397**	**0.091746**	**-7.906570**	**0.0000**
C	-0.093795	0.259164	-0.361913	0.7176
AO	-30.830498	5.565333	-5.539740	0.0000

R-squared	0.246172	Mean dependent var	-0.089549
Adjusted R-squared	0.236749	S.D. dependent var	7.819207
S.E. of regression	6.831189	Akaike info criterion	6.698757
Sum squared resid	22399.271496	**Schwarz criterion**	**6.758958**
Log likelihood	-1624.147318	Hannan-Quinn criter.	6.722406
Jarque-Bera (Prob.)	24.237 (0.0000)	Durbin-Watson stat	1.997969

Inverted AR Roots	0.97 0.94 0.47+/-0.82i 0.82+/-0.47i
	-0.47+/-0.82i 0.00+/-0.94i -0.94 -0.82+/-0.47i
Inverted MA Roots	0.84+/-0.49i 0.00+/-0.97i -0.84+/-0.49i -0.97
	-0.49+/-0.84i 0.49+/-0.84i 0.97 0.89
	0.51

Figura 20: Modelo de la serie desestacionalizada (comparación, regla 8).